

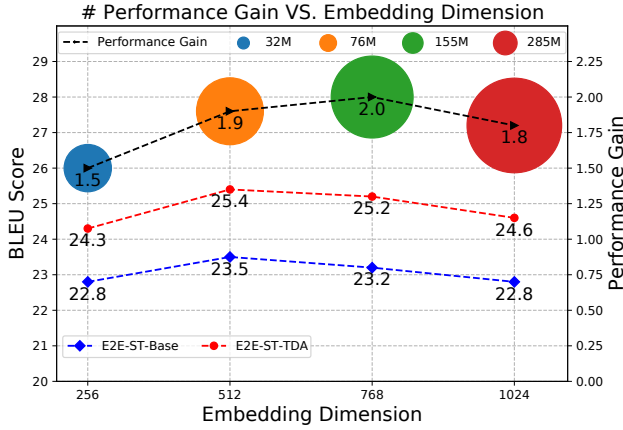


Figure 3: The impact of model size on performance. The area of the circle represents the parameter size of the model.

a smaller λ will make the model pay more attention to optimizing the dual-path. In the early stage of model training, since the learned parameters are not perfect, a larger λ will cause the parameters of the model to be constrained in the wrong position by agreement earlier, making it challenging to further optimize the model. However, a smaller λ will make the dual-path too independent, and it is not easy to narrow the output representation.

Related Work

Speech Translation. Early ST methods (????) cascade the ASR subsystem and the MT subsystem. With the rapid development of deep learning, the neural networks widely used in ASR and MT have been adapted to construct a new end-to-end speech-to-text translation paradigm. However, due to the scarcity of triplet training data, developing an E2E-ST model that does not use intermediate transcription is still very challenging. Thus, various techniques have been proposed to ease the training process by using source transcriptions, including pre-training (???), multi-task learning (???), meta-learning (?), interactive decoding (?), consecutive decoding (?), and adapter tuning (?). Among them, the pre-training strategy is a simple and effective way that pre-train different components of the ST system and merges them into one. However, the training process of these methods only loosely couples the two type two-tuple data. It does not fully explore the potential connections between the triplet data. Therefore, how to fully mining the associations between the scarce triplet data still remains a crucial issue to improve the performance of E2E-ST. Following this research line, we introduce TDA to fully exploit the association relationship in triplet data.

Agreement Regularization. One line of agreement regularization attempts to regularize model predictions to be invariant with minute perturbations on input data, which focused on semi-supervised learning areas. The minute perturbations can be random noise (?), adversarial noise (????), gaussian noise (?) and various data augmentation meth-

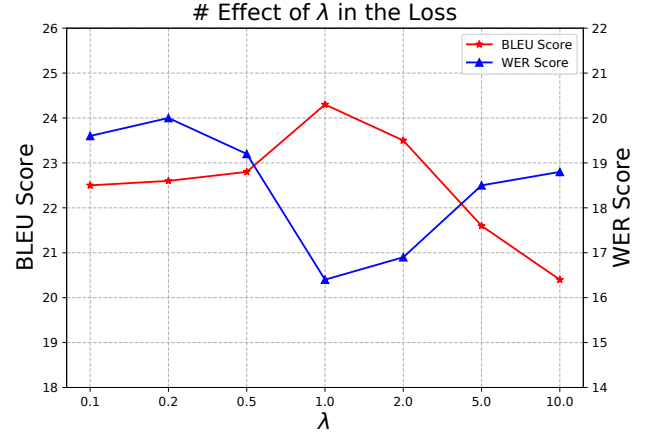


Figure 4: The impact of the hyper-parameter λ in the loss function on performance.

ods (??). Another line tries to take into consideration the agreement between the different models, especially in sequence modeling. For instance, there are some attempts in speech recognition (?), neural machine translation (??), and speech synthesis (?), which try to improve the performance by integrating the predicted probability from forward and backward decoding sequences. Our method is most similar to the latter, but we aim to constrain the agreement between the probability distributions of two directions sequences in a single unified model.

Conclusion

In this paper, we propose a flexible and effective regularization method for the ST task, namely Triangular Decomposition Agreement (TDA), which relies on the agreement between inherent and unexplored dual decomposition paths ASR-MT and ST-BT of ST. In our method, two Kullback-Leibler divergences are added to the standard training objective as regularization terms to resolve the mismatch between the joint probability distributions of dual-path ASR-MT and ST-BT. In addition, our approach can use two paths for inference and adopt early termination methods for different scenarios to ensure efficient inference speed. Empirical evaluations of the eight translation directions of MuST-C demonstrate that our proposed approach leads to significant improvements compared with strong baseline systems.

Acknowledgements

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